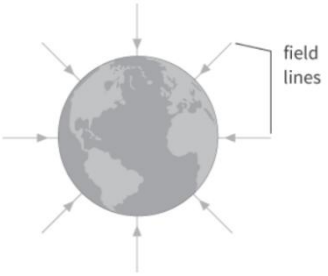
13.2.1 Exercises - Gravitational Field 1

1. This diagram shows how we can represent the gravitational field of the Earth:



a. Explain why the arrows on the field lines are all directed into the Earth.

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b. At what point do all the field lines appear to meet? The building you are sitting in is on the surface of the Earth. Here, the Earth's gravitational field is described as uniform.

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c. Draw a diagram to show gravitational field lines in the area around you.

d. State how the field lines you have drawn show that the field is uniform.

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e. Look back at Figure. Explain how this shows that the Earth's gravitational field is not uniform on this scale.

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2. This diagram shows an object of mass Mon the surface of the Earth:

a. Add an arrow to show the direction of the Earth's gravitational force F on the object.

b. The object is attracted to all points within the Earth. Which region of the Earth causes the strongest gravitational pull on the object? Label this region A. Explain your answer.

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c. Which region of the Earth causes the weakest gravitational pull on the object? **Label this region B**. Explain your answer.

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d. We can consider the Earth's gravitational pull as if all of its mass were concentrated at one point, its centre of gravity. **Label this point C**. Explain your answer.

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3. Partly because the Earth is not a perfect sphere,the value of g varies from 9.78 Nkg^{-1} at the equator to 9.83 Nkg^{-1} at the poles.

An object of mass 20.0 kg is transported from the equator to the North Pole.

a. Calculate the weight of an object of mass 20.0 kg at a point where the Earth's gravitational field strength g=9.80 Nkg^{-1}

b. Calculate the amount by which the object's weight increases in moving from the equator to the North Pole.

c. Explain how g will change if you climb to the top of a high mountain.

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4. The gravitational force of one mass on another is given by Newton's law, The definition of gravitational field strength g says that its value is the gravitational force acting per unit mass on a body.

a. Express the gravitational field strength at a distance r from a point mass M:

b. Calculate the gravitational field strength at the Earth's surface. Give your answer to two significant figures. (radius of Earth = 6400 km)

5. The Moon has a mass of 7.4×10^{22}kg and its mean radius is 1.74×10^{6}m.

a. Calculate the gravitational force on a mass of exactly 1 kg on the surface of the Moon.

b. State the value of the gravitational field strength on the surface of the Moon.

c. Calculate the weight of an object of mass 20.0 kg on the surface of the Moon.

d. Imagine that you dropped an object close to the surface of the Moon. Determine the object's acceleration.

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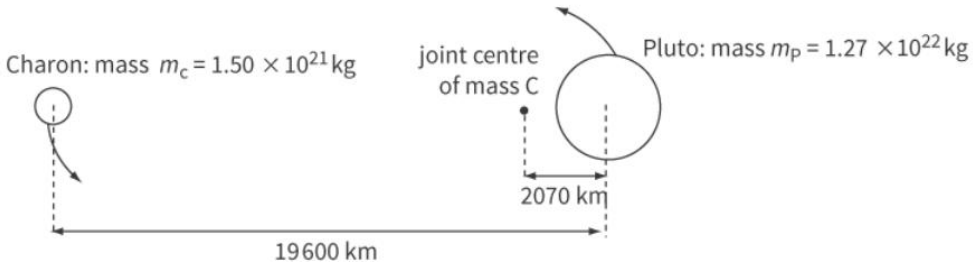
6. The dwarf planet Pluto is orbited by its moon Charon,as shown:

In fact they orbit one another around their joint centre of mass C.



a. Calculate the gravitational field strength at C due to Pluto. Give its direction

b. Calculate the gravitational field strength at C due to Charon. Give its direction.

c. What is the gravitational force on an object of mass 1.0 kg placed at C? Explain your answer.

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7. The Earth has a mass of 5.98×10^{24} kg and a radius of 6.37×10^{6} m.
The Moon has a mass of 7.35×10^{22} kg and a radius of 1.74×10^{6} m.
The Earth and the Moon can both be considered as point masses at their centres. Their centres are a distance of 3.84×10^{8} m apart.

a. Show that the gravitational field strength at the surface of the Moon due to the mass of the Moon is 1.62 Nkg^{-1}. [1]

b. Explain why there is a point X on the line between the centres of the Earth and the Moon where the resultant gravitational field strength due to the Earth and the Moon is zero.

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c. Calculate the distance x of point X from the centre of the Moon.

x = ... m [3]